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SGD without Variance Assumption

New Tight Bounds via a Computer-Aided Lyapunov Analysis

Daniel Cortild, Lucas Ketels, J. Peypouquet, G. Garrigos

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Stochastic Gradient Descent

Consider the problem

$$\min \left\{ f(x) = \frac{1}{n} \sum_{i=1}^n f_i(x) : x \in \mathbb{R}^d \right\},$$

where all $f_i: \mathbb{R}^d \rightarrow \mathbb{R}$ are convex and L -smooth, and f has minimizers.

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Stochastic Gradient Descent (SGD) iterates

$$x_0 \in \mathbb{R}^d, \quad x_{t+1} = x_t - \gamma \nabla f_{i_k}(x_t) \quad \text{for } t = 0, 1, \dots,$$

where i_k is chosen i.i.d. from the uniform distribution on $\{1, \dots, n\}$.

Type of Results for SGD

Convergence results for SGD are usually presented as

$$\text{Performance}(t) \leq \text{Bias}(t) + \text{Variance}(t),$$

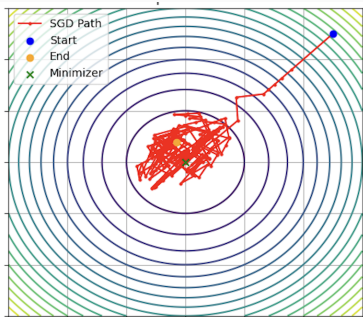
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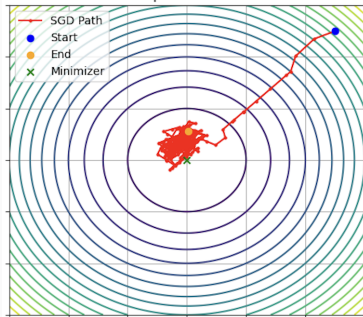
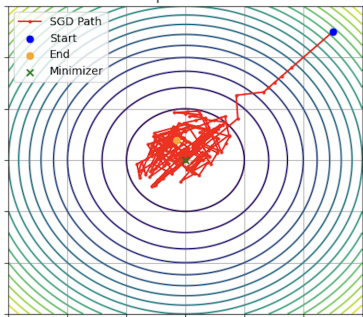


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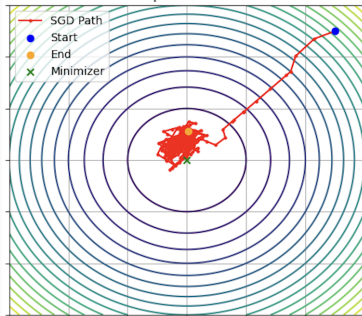
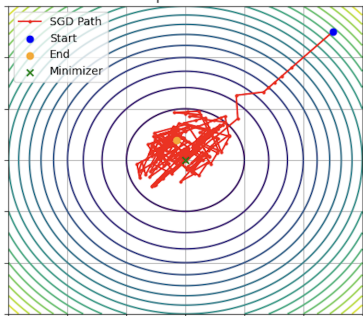


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Goal: Minimize the bias term first, and the variance term second.

Variance Assumption

¹Bottou, Curtis, and Nocedal, “Optimization Methods for Large-Scale Machine Learning”, 2018.

Typical Assumption: Uniformly bounded gradient variance;

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}[\|\nabla f_{i_k}(x) - \nabla f(x)\|^2] < +\infty.$$

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Our setting: We define

$$\sigma_*^2 := \mathbb{E}[\|\nabla f_{i_k}(x_*)\|^2] \quad \text{for some } x_* \in \operatorname{argmin} f.$$

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Note this is automatically finite in our setting.

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Results in Convex Setting

We obtain a result on the Cesàro average $\bar{x}_T = \frac{x_0 + \dots + x_{T-1}}{T}$ of the form

$$\mathbb{E}[f(\bar{x}_T) - \min f] \leq \text{Bias}(T) \cdot \|x_0 - x_*\|^2 + \text{Variance}(T) \cdot \sigma_*^2,$$

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$$\text{Bias}(T) = \begin{cases} \frac{1}{2\gamma T} & \text{if } \gamma L \in (0, 1), \\ \frac{1}{(2-\varepsilon)\gamma T} & \text{if } \gamma L = 1, \varepsilon > 0, \\ \frac{1}{2\gamma(2-\gamma L)T} & \text{if } \gamma L \in (1, 2), \end{cases}$$

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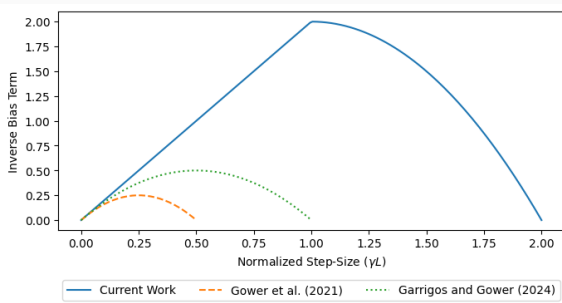
Observation 2: No uniform bound in T for $\gamma L > 1$.

Observation 3: If $\sigma_*^2 = 0$, these are not problems.

Comparison to State-of-the-Art

Comparison to

- Gower et al. (2021)²,
- Garrigos and Gower (2024)³.



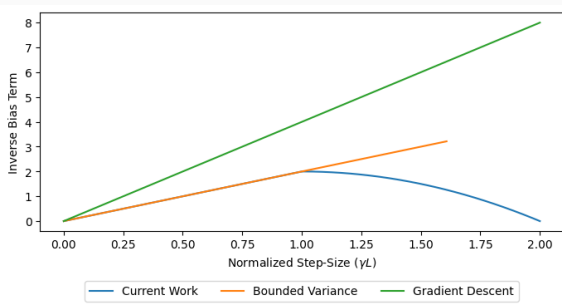
²Gower, Sebbouh, and Loizou, “SGD for Structured Nonconvex Functions: Learning Rates, Minibatching and Interpolation”, 2021.

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Comparison to Other Methods

Comparison to

- SGD with Uniformly Bounded Variance⁴,
- Gradient Descent⁵.



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Proof Strategy 1/3

Our proofs are based on a Lyapunov analysis with an energy of the form

$$E_t := a_t \|x_t - x_*\|^2 + \rho \sum_{s=0}^{t-1} [f(x_s) - \min f] - \sum_{s=0}^{t-1} e_s \sigma_*^2,$$

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We then aim to minimize $\text{Bias}(T) = \frac{a_0}{\rho T}$.

Obtaining the best bias may then be formulated as

$$\text{Bias}_{\text{opt}}(T) = \inf \{ \text{Bias}(T) : (a_t), (e_t), \rho \text{ are Lyapunov parameters} \}.$$

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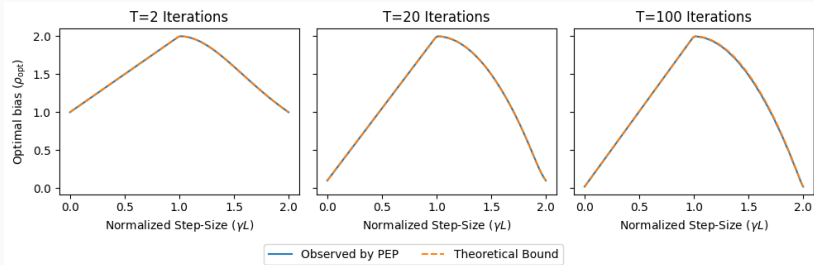
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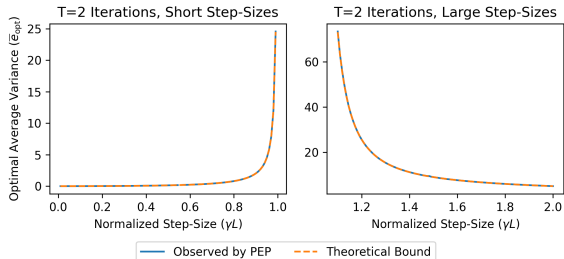
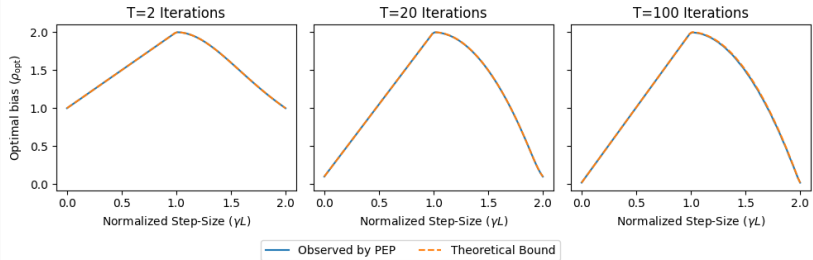
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Which helped us getting a *theoretical bias* term $\text{Bias}_{\text{theory}}(T)$.

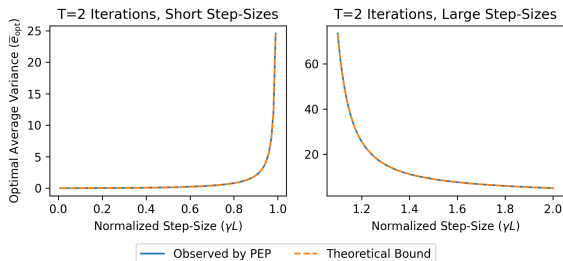
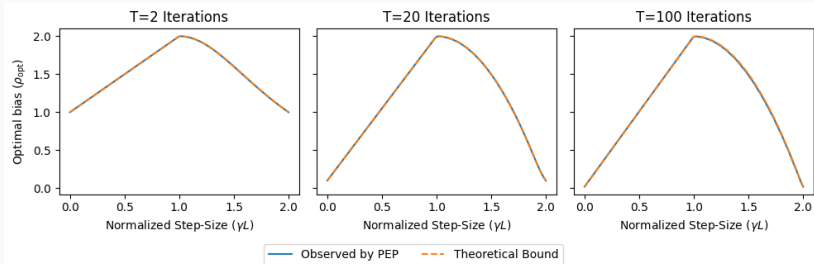
Tightness of Bias and Variance Terms



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Note: We only claim tightness within our framework.

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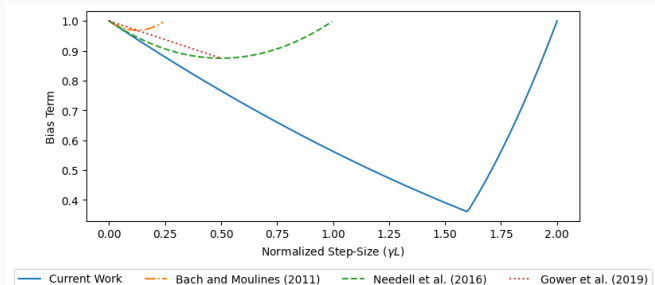
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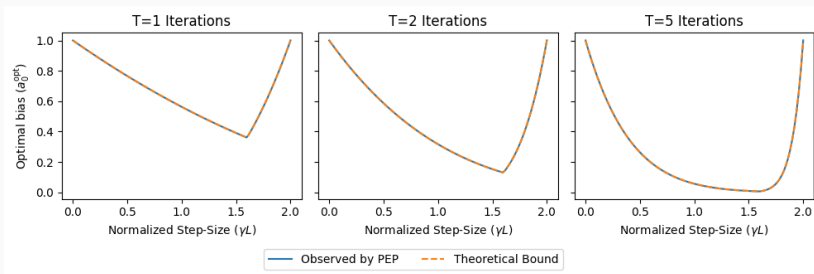
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Comparison to state-of-the-art⁹:

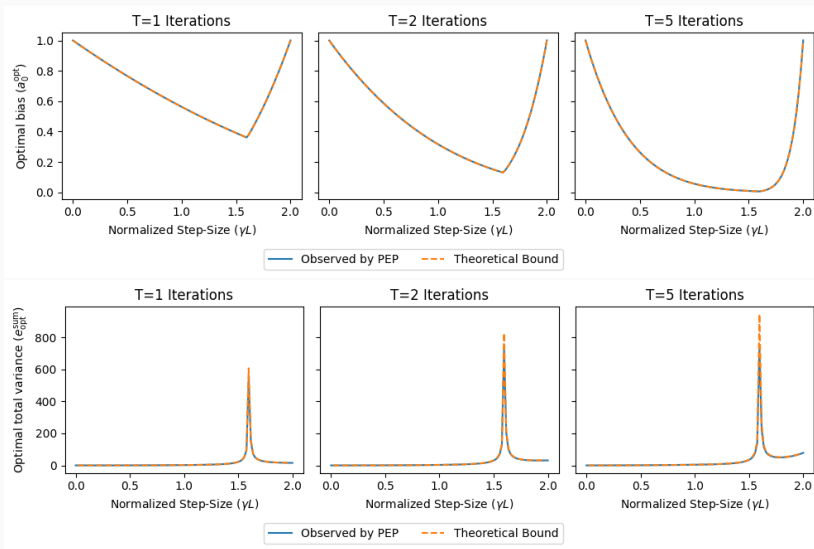


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Thank you!

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We obtain a bound of the form

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where

$$\text{Bias}(T) = \left(\max\{1 - \gamma\mu, \gamma L - 1\}^2 + \varepsilon \right)^T,$$

and

$$\text{Variance}(T) = \mathcal{O} \left(\gamma^2 + \frac{\gamma^4}{\varepsilon + \left| \gamma - \frac{2}{L+\mu} \right|} \right).$$

Observation 1: If $\varepsilon \neq 0$, the bound is valid for all $\gamma > 0$, and the bias goes to 0 if $\gamma L \in (0, 2)$.

Observation 2: If $\gamma \neq \frac{2}{L+\mu}$, we can take $\varepsilon = 0$. This is in analogy with the singularity observed earlier at the optimal step-size.